

Apex Inventory

UI Requirement Specification

Dashboard Overview

Route: /

Document Version: 1.0.0

Project: ReactExample (Apex Store Inventory Management)

1. Overview

The Dashboard is the default landing page (route '/'). It provides a real-time operational overview of the store inventory: KPI metrics, stock alerts, upcoming deliveries, and analytics charts for sales trends, product demand comparison, and seasonal performance.

2. Purpose

- Give managers an at-a-glance view of inventory health.
- Surface urgent stock issues (out of stock, low stock).
- Show replenishment orders arriving within the next 7 days.
- Visualize sales demand trends and seasonal patterns for planning.

3. Page Sections

3.1 Page Header

Title: 'Dashboard Overview'. Subtitle: 'Real-time analytics, stock warnings, and seasonal demand intelligence'.

3.2 KPI Cards (4 cards in a grid)

KPI	Calculation
Total Unique Items	Count of all inventory items
Out of Stock	Items where quantity === 0
Low Stock (1-5)	Items where 0 < quantity <= 5

Pending Orders	Orders with status === 'pending'
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3.3 Stock Alert Center (Left panel)

Two sub-sections:

- Out of Stock (0 units): Lists items with quantity 0. Shows name, category, 'Out of Stock' badge. Empty state: 'No items are currently out of stock. Excellent!'
- Low Stock Warning (1 to 5 units): Lists items with quantity 1-5. Shows name, category, quantity badge. Empty state: 'No low stock items. Inventory healthy!'
- Header badge shows total alert count (out of stock + low stock).

3.4 Stocking Up - Coming Week (Right panel)

Lists pending orders with expectedArrival between today and 7 days from now. Each entry shows: item name, arrival date, quantity (+N), physical state. Empty state: 'No arrivals this week' with truck icon.

3.5 Sales Demand Trend Chart

Area chart (Recharts) showing 6-month sales trend (Jan-Jun) for a selected item. Dropdown selector lists all inventory items; defaults to first item. Data source: item.salesTrend array (6 values). Y-axis: sales volume. Interactive tooltip on hover.

3.6 Product Demand Comparison Chart

Bar chart comparing total demand across all items. Total demand = sum of salesTrend array. Items sorted descending by demand. Item names truncated to 15 chars if longer.

3.7 Seasonal Product Performance Chart

Grouped bar chart across four seasons: Summer, Monsoon, Winter, Spring. Shows up to first 5 items as separate colored bars per season. Data source: item.seasonalSales object. Legend displays item names.

4. Data Loading

On mount, fetches GET /api/items and GET /api/orders in parallel via Promise.all. Errors logged to console; page renders with empty data if fetch fails. selectedTrendItemId auto-sets to first item ID when items load.

5. User Interactions

- Item selector dropdown: changes Sales Demand Trend chart to selected item.
- Chart tooltips: hover to see exact values (Recharts built-in).
- No edit/delete actions on Dashboard - read-only analytics view.

6. Business Rules

- Low stock threshold: quantity <= 5
- Out of stock: quantity === 0

- Upcoming arrivals: pending orders only, arrival date within next 7 calendar days
- Date comparisons use JavaScript Date objects

7. Technical Notes

- Component: src/pages/Dashboard.tsx
- Chart library: recharts (AreaChart, BarChart, ResponsiveContainer)
- Icons: lucide-react